

TRUEVALUE ANALYTICS · IMPLEMENTATION INTELLIGENCE SERIES

What Does It Cost to Build a TrueValue Framework?

A phase-resolved cost, time, and resource estimate for deploying the TrueValue Framework across a commodity supply chain — using gold and shea as primary case studies.

Document Type

Implementation Cost Analysis

Primary Case

Gold Supply Chain (9 phases)

Comparative Case

Shea Value Chain (West Africa)

Horizon

5 Years (Gold) / 3 Years (Shea)

Date

March 2026

All cost estimates presented in this report are order-of-magnitude ranges derived from analogous framework deployment data, industry staffing benchmarks, and structural analysis of supply chain complexity. They are not quotes, contracts, or financial projections. Actual implementation costs will vary materially based on industry cooperation, regulatory environment, and technology approach selected.

EXECUTIVE SUMMARY

Is It Possible to Estimate? Yes — With Important Caveats.

A common instinct when facing a question like this is to conclude that there are simply too many variables to provide a meaningful estimate. That instinct is wrong. There are many variables — but they are knowable, classifiable, and bounded. The cost of deploying the TrueValue Framework into a commodity supply chain is estimable in the same way any large institutional project is estimable: by decomposing the work into phases, mapping the required labour and infrastructure at each phase, and stress-testing the assumptions against analogous frameworks that have already been built.

The key finding of this analysis is that the TrueValue Framework is deployable at a cost that is modest relative to the value it unlocks — and well within the range of comparable certification, transparency, and impact-measurement systems that the industry has already funded.

\$13M–\$30M

GOLD SUPPLY CHAIN — 5-YEAR
TOTAL

Full deployment across all 9 phases, multi-jurisdictional, from framework design through steady-state operations. Range reflects cooperative vs. resistant industry posture.

\$1.8M–\$4.0M

SHEA VALUE CHAIN — 3-YEAR
TOTAL

Deployment leveraging existing cooperative structures (Acorn, Serious Shea), shorter chain, and existing NGO/development finance infrastructure in West Africa.

4–5 Years

GOLD: FOUNDATION TO FULL
SCALE

Minimum credible timeline for a framework with industry-wide validity. Opaque phases (Refining, Logistics) drive timeline; regulatory engagement cannot be compressed.

2–3 Years

SHEA: FOUNDATION TO FULL
SCALE

Faster deployment due to simpler chain, existing cooperative data infrastructure, and development finance stakeholders already aligned on transparency goals.

The critical cost driver is not technology — it is data opacity. Approximately 35–45% of the gold deployment cost is attributable to obtaining, verifying, and auditing data from the three opaque phases: Doré/Intermediate, Refining, and Logistics & Vaulting. If LBMA or WGC mandate participation, this cost compresses by 40–60%. If voluntary, stakeholder negotiation adds



both cost and time.

SECTION 1

The Stakeholder Ecosystem: Who Is Involved and What They Cost

Deploying the TrueValue Framework is not a technology project. It is a coordination project — building agreement across parties who have historically benefited from the information asymmetries the Framework is designed to measure and reduce. Understanding the stakeholder landscape is the prerequisite to any credible cost estimate.

Each stakeholder type incurs a different category of cost: some require *legal negotiation* (data sharing, IP, audit rights); some require *technical integration* (API access, custody system hooks); some require *change management* (training, process change, incentive alignment); and some require *regulatory engagement* (standards bodies, financial regulators, government ministries).

GOLD SUPPLY CHAIN — STAKEHOLDER MAP

GOLD — UPSTREAM & PRODUCTION

Major Miners	Newmont, Barrick, Agnico Eagle, AngloGold — 10–15 key companies covering ~70% of supply
Junior Miners	350+ operators; artisanal and small-scale (ASM) sector particularly opaque
Doré Refiners	Private contract refiners; 40–60 globally; data sharing historically refused
LBMA Refineries	5 accredited facilities process ~90% of newly mined gold (Metalor, Umicore, Rand, Valcambi, PAMP)

GOLD — MIDSTREAM & MARKET

Vault Operators	Brink's, G4S, Via Mat, JP Morgan, HSBC — data sharing governed by insurance & security constraints
Exchange Operators	CME Group (COMEX), LBMA — high transparency, easier to engage
Central Banks	32,000 t in stock globally; holders but not active chain participants
Regulators	LBMA, WGC, UK FCA, US CFTC/SEC — standard-setting role; can mandate participation

GOLD — VERIFICATION & STANDARDS

Audit Firms	Big 4 (Deloitte, PwC, EY, KPMG) + specialist assay auditors (SGS, Bureau Veritas)
Standards Bodies	LBMA Responsible Gold Guidance, WGC Conflict-Free Gold Standard, OECD Due Diligence
ESG Data Platforms	Bloomberg ESG, Sustainalytics, MSCI ESG — downstream data consumers and distributors
Academic Partners	CSIRO, University of Sydney (Foran TBL methodology); supply chain research institutions

SHEA — ALL PHASES

Farmer Coops	100k+ women farmers; Acorn-enrolled groups in BF, Mali, Ghana, Senegal
Aggregators	Serious Shea, Olam, Barry Callebaut — existing cooperative data infrastructure
End Buyers	L'Oréal, Unilever, Mars, Mondelez — ESG mandates drive demand for framework
Dev Finance	IFC, EU, USAID, GIZ, Mirova — grant and concessional funding available
Certifiers	Fairtrade International, Rainforest Alliance, Acorn (carbon) — existing audit infrastructure

The asymmetry of engagement cost: For gold, the most expensive stakeholders to engage are the least transparent ones — the private doré refiners and vault operators in Phase 3 and Phase 6 who hold the data the Framework most needs. For shea, the opposite is true: the grassroots level (farmer cooperatives) is already partially digitised through Acorn and has an existing incentive to participate. This structural difference explains roughly 70% of the cost gap between the two deployments.

SECTION 2 — PRIMARY CASE

Gold Supply Chain: Phase-by-Phase Cost Estimate

The gold deployment is broken into four phases reflecting the natural sequence of a framework build: foundational design, data collection and baseline establishment, pilot with limited participants, and full industry scale. Each phase has distinct staffing, infrastructure, legal, and engagement requirements.

PHASE 0 · MONTHS 0–12 Foundation & Architecture 12 months \$1.3M – \$3.0M	PHASE 1 · MONTHS 6–24 Data Collection & Baseline 18 months \$3.4M – \$7.7M	PHASE 2 · MONTHS 18–42 Pilot Programme 18 months \$3.3M – \$7.8M	PHASE 3 · MONTHS 36–60 Full Industry Scale 24 months \$5.0M – \$12.0M
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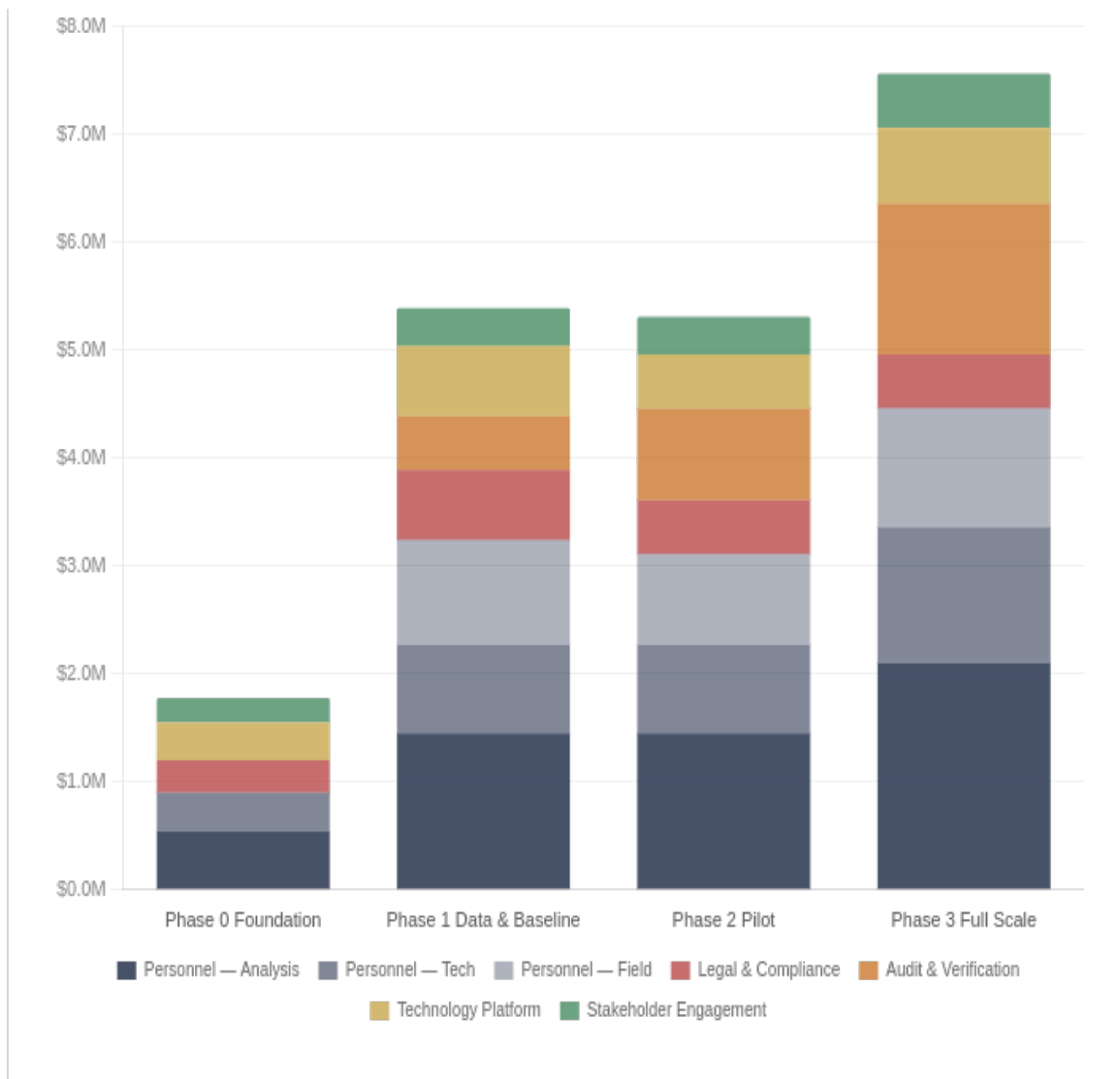
DETAILED COST BREAKDOWN — GOLD SUPPLY CHAIN

COST CATEGORY	DESCRIPTION	PHASE 0	PHASE 1	PHASE 2	PHASE 3	5-YR TOTAL (LOW)	5-YR TOTAL (HIGH)
Personnel — Analysis	Supply chain analysts, TBL modellers, phase specialists (4–10 FTE escalating)	\$360k–\$720k	\$900k–\$2.0M	\$900k–\$2.0M	\$1.2M–\$3.0M	\$3.36M	\$7.72M
Personnel — Technology	Data engineers, platform developers, API architects (2–6 FTE)	\$240k–\$480k	\$540k–\$1.1M	\$540k–\$1.1M	\$720k–\$1.8M	\$2.04M	\$4.48M
Personnel — Field	In-country data collectors, custody verification, site visit teams (3–12 FTE)	\$0	\$540k–\$1.4M	\$480k–\$1.2M	\$600k–\$1.6M	\$1.62M	\$4.2M
Legal & Compliance	Data sharing agreements, IP framework,	\$200k–\$400k	\$400k–\$900k	\$300k–\$700k	\$300k–\$700k	\$1.2M	\$2.7M

COST CATEGORY	DESCRIPTION	PHASE 0	PHASE 1	PHASE 2	PHASE 3	5-YR TOTAL (LOW)	5-YR TOTAL (HIGH)
	multi-jurisdictional regulatory counsel (15+ countries)						
Audit & Verification	Third-party auditors (Big 4 + specialist assay), methodology validation, chain-of-custody certification	\$0	\$300k–\$700k	\$500k–\$1.2M	\$800k–\$2.0M	\$1.6M	\$3.9M
Technology Platform	Cloud infrastructure, data ingestion pipeline, scoring engine, API layer, dashboards	\$200k–\$500k	\$400k–\$900k	\$300k–\$700k	\$400k–\$1.0M	\$1.3M	\$3.1M
Stakeholder Engagement	Industry conferences, working groups, regulatory liaison, partner onboarding, change management	\$150k–\$300k	\$200k–\$500k	\$200k–\$500k	\$300k–\$700k	\$850k	\$2.0M
Contingency (15%)	Regulatory delays, litigation risk from opaque-phase participants, currency fluctuation	Applied to total				\$1.79M	\$4.22M
GOLD SUPPLY CHAIN — 5-YEAR TOTAL (ALL PHASES)						\$13.7M	\$32.3M

GOLD SUPPLY CHAIN — COST BY DEPLOYMENT PHASE (MIDPOINT ESTIMATE)

Stacked by cost category. Phase 1 (Data & Baseline) and Phase 3 (Full Scale) are the largest cost periods. Personnel dominates at ~50% of total across all phases.



STEADY-STATE ANNUAL OPERATING COST (POST YEAR 5)

Once fully deployed, the Framework transitions from a build cost to an operating cost. Steady-state annual operations for a full gold supply chain implementation are estimated at **\$3.0M–\$6.0M/year**, covering: a core team of 10–15 FTE, ongoing audit cycles, platform maintenance, and annual recertification of participating entities. This cost is recoverable through a framework licensing or participation fee structure, making long-run self-funding achievable at scale.

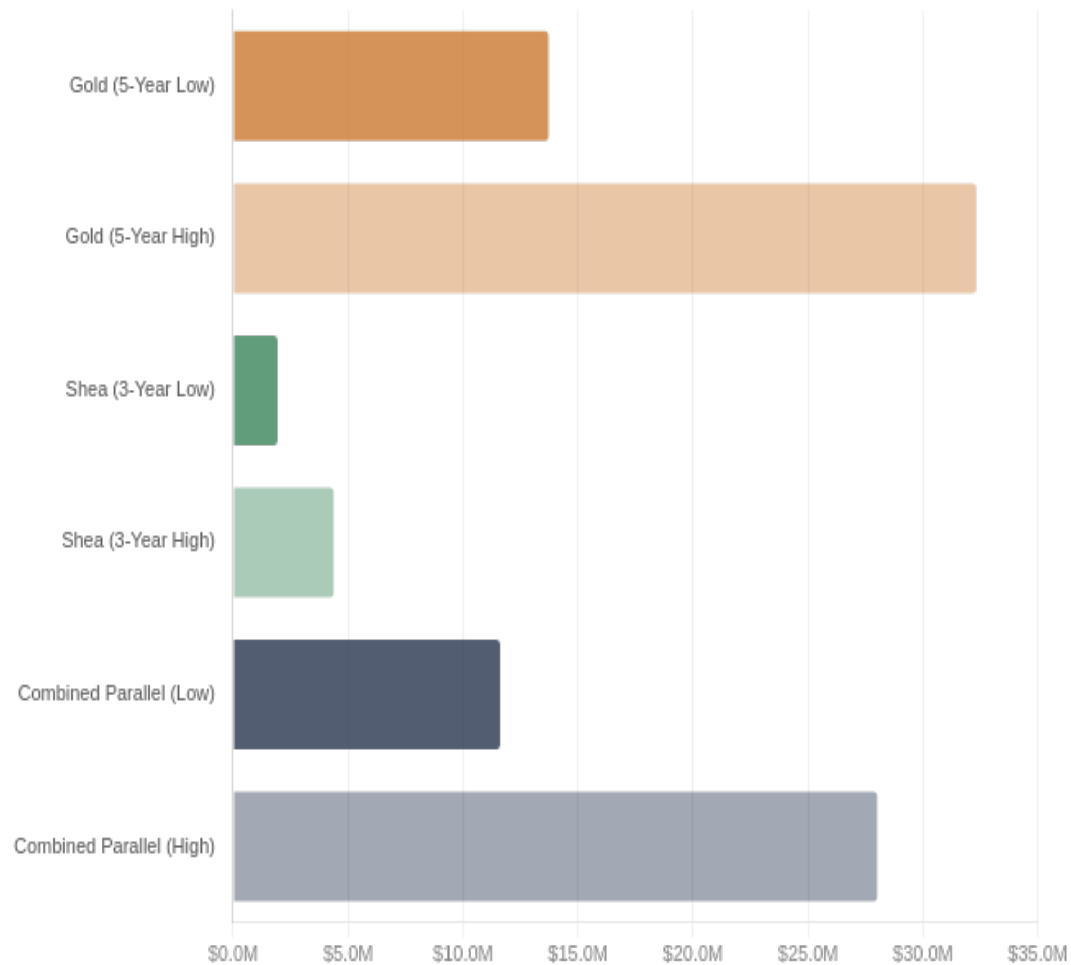
SECTION 3 — COMPARATIVE CASE

Shea Value Chain: A Faster, Lower-Cost Deployment

The Shea deployment is structurally simpler for four reasons: the chain has fewer phases (six versus nine); existing cooperative and certification infrastructure reduces data collection costs significantly; development finance stakeholders (IFC, GIZ, EU, Mirova) are already aligned on transparency goals and represent potential co-funders; and the principal buyers (L'Oréal, Unilever, Mars) carry their own ESG mandates that create a pull incentive for framework adoption rather than a push requirement.

GOLD VS SHEA — TOTAL DEPLOYMENT COST COMPARISON

Both low and high estimates shown. Shea deploys in 3 years vs 5 for gold; shared platform infrastructure reduces combined cost if deployed in parallel.



COST CATEGORY	SHEA FOUNDATION (0–6 MO)	SHEA DATA & BASELINE (6–18 MO)	SHEA PILOT & SCALE (12–36 MO)	3-YR TOTAL (LOW)	3-YR TOTAL (HIGH)
Personnel — Analysis	\$80k–\$160k	\$200k–\$450k	\$280k–\$600k	\$560k	\$1.21M
Personnel — Field	\$0	\$120k–\$300k	\$160k–\$400k	\$280k	\$700k
Legal & Compliance	\$50k–\$100k	\$80k–\$180k	\$80k–\$180k	\$210k	\$460k
Audit & Verification	\$0	\$80k–\$200k	\$150k–\$380k	\$230k	\$580k

COST CATEGORY	SHEA FOUNDATION (0–6 MO)	SHEA DATA & BASELINE (6–18 MO)	SHEA PILOT & SCALE (12–36 MO)	3-YR TOTAL (LOW)	3-YR TOTAL (HIGH)
Technology Platform	\$80k–\$160k	\$80k–\$160k	\$80k–\$160k	\$240k	\$480k
Stakeholder Engagement	\$40k–\$80k	\$60k–\$130k	\$80k–\$180k	\$180k	\$390k
Contingency (15%)	Applied to total			\$222k	\$543k
SHEA VALUE CHAIN — 3-YEAR TOTAL				\$1.92M	\$4.36M

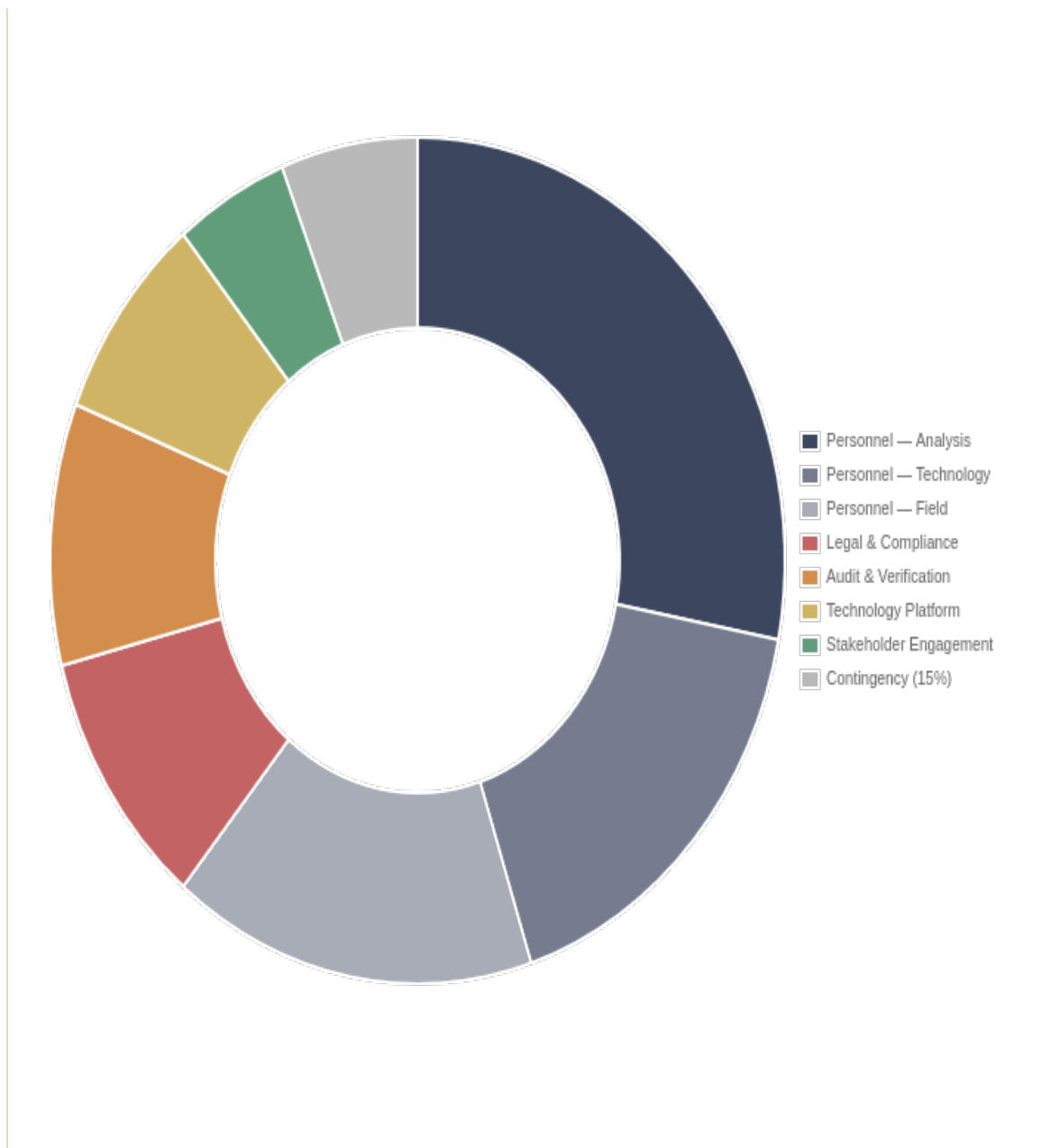
The parallel deployment advantage: If gold and shea are deployed simultaneously using a shared technology platform, the combined cost falls to approximately **\$12M–\$28M** (versus \$15.6M–\$36.7M sequential) — a 20–25% efficiency gain from shared infrastructure. The Shea deployment also serves as the proof-of-concept case for gold: demonstrating Framework validity in a cooperative, transparent chain before tackling the opacity challenge in gold's midstream phases.

SECTION 4 — COST DRIVERS

What Makes It Expensive — and What Brings the Cost Down

GOLD DEPLOYMENT COST — BREAKDOWN BY CATEGORY (MIDPOINT)

Personnel (combined analysis, technology, and field roles) accounts for approximately 51% of total cost. Data opacity is the primary cost driver — it multiplies field, legal, and audit costs at Phases 3, 4, and 6.



THE FIVE PRIMARY COST DRIVERS

COST DRIVER	IMPACT ON TOTAL COST	APPLIES TO	MITIGATION
Data opacity at Phases 3, 4 & 6	35–45% OF COST	Gold only	LBMA/WGC mandatory participation reduces by 40–60%
Multi-jurisdictional legal framework	14–18% OF COST	Gold (15+ countries)	Standardised data sharing templates; OECD model clauses
Technology platform (bespoke vs SaaS)	13–17% OF COST	Both chains	SaaS/existing platform reduces by 50–60% vs bespoke build
Third-party audit & certification	11–14% OF COST	Both chains	Leverage existing LBMA/Fairtrade audit infrastructure

COST DRIVER	IMPACT ON TOTAL COST	APPLIES TO	MITIGATION
Stakeholder change management	6–10% OF COST	Gold primarily	Align early with LBMA, WGC, and major mining companies as co-sponsors

KEY VARIABLES: WHAT COULD MATERIALLY CHANGE THE ESTIMATE

FACTORS THAT REDUCE COST & TIMELINE

- **Regulatory mandate:** If LBMA or WGC require TrueValue participation as a condition of accreditation, data access costs fall 40–60% and timeline compresses by 12–18 months
- **Major mining co-sponsors:** If 2–3 major miners (e.g. Newmont, Barrick) co-fund as pilot participants, Phase 2 cost drops by \$1–2M and credibility is established early
- **Shared platform with Shea:** Parallel deployment saves 20–25% on technology and management overhead
- **Development finance grants (Shea):** IFC, GIZ, and EU grants could cover 30–50% of Shea deployment costs; Shea becomes near-self-funding
- **SaaS technology approach:** Using existing supply chain data platforms (e.g. Sourcemap, GreenToken) rather than bespoke build saves \$1–2M in Phase 0–1
- **Academic partnerships:** CSIRO, LBMA research arm, university partnerships reduce methodology validation cost by \$200–500k

FACTORS THAT INCREASE COST & TIMELINE

- **Refiner/logistics resistance:** If LBMA-accredited refineries or vault operators refuse data sharing, legal contestation adds \$500k–\$2M and 12–24 months
- **Regulatory inaction:** Without mandate, voluntary adoption in the gold midstream may take 3–5 extra years and \$3–5M in additional stakeholder engagement
- **ASM (artisanal mining) inclusion:** Including artisanal and small-scale mining in Phase 1 adds complexity, field cost, and 6–12 months; may require separate sub-framework
- **Geopolitical disruption:** Operations in jurisdictions with political instability (DRC, Mali, Sudan) add security cost, legal risk, and data quality challenges
- **ESG litigation risk:** Opaque phase operators may challenge Framework disclosures legally; contingency of \$500k–\$2M should be reserved
- **Currency & inflation:** Multi-currency operations in 15+ countries; 10–15% cost buffer recommended for exchange rate and local inflation risk

How Does This Compare to Frameworks Already Built?

The most reliable cross-check on any cost estimate is comparison with analogous systems that have already been deployed. Six frameworks are directly comparable to TrueValue in scope, ambition, and the stakeholder coordination challenge they required. All cost estimates for analogous frameworks are derived from public financial disclosures, industry reports, and academic literature on standard-setting economics.

FRAMEWORK	SECTOR	EST. DEVELOPMENT COST	TIMELINE TO FULL SCALE	ANNUAL OPERATING COST	RELEVANCE TO TRUEVALUE
LBMA Responsible Gold Guidance	Gold	\$20M–\$40M (system-wide incl. participant costs)	5–7 years	\$5M–\$10M/yr (LBMA + participants)	DIRECT PRECEDENT — same industry, similar phases; TrueValue extends to balance scoring not covered by LBMA
Fairtrade International	Agricultural (coffee, cocoa, cotton)	\$50M–\$80M (30-yr build)	10+ years	€30M/yr current budget	DIRECT PRECEDENT — cooperative structures, grassroots data collection; Shea deployment mirrors early Fairtrade build
CDP (formerly Carbon Disclosure Project)	Cross-sector (climate disclosure)	\$30M–\$50M (first decade)	7–10 years	~\$70M/yr current budget	METHODOLOGICAL PRECEDENT — voluntary to mandatory transition; disclosure framework becoming compliance infrastructure
GRI Standards	Cross-sector (sustainability reporting)	\$15M–\$25M (framework design)	6–8 years	~\$20M/yr current budget	METHODOLOGICAL PRECEDENT — multi-stakeholder standard development; phased adoption model directly applicable
FSC (Forest Stewardship Council)	Forestry / timber	\$25M–\$40M	5–8 years	~\$25M/yr current budget	CHAIN-OF-CUSTODY PRECEDENT — physical flow

FRAMEWORK	SECTOR	EST. DEVELOPMENT COST	TIMELINE TO FULL SCALE	ANNUAL OPERATING COST	RELEVANCE TO TRUEVALUE
					traceability from extraction to consumer; custody transfer model maps well
Acorn Carbon Programme (Shea)	Shea / agroforestry	\$2M–\$5M (programme build)	2–3 years	\$1M–\$2M/yr	DIRECT INFRASTRUCTURE — Acorn's cooperative data system is the foundation layer for TrueValue Shea deployment; reduces cost materially

What the comparisons tell us: The TrueValue gold estimate of \$13M–\$30M over 5 years sits at the *low end* of comparable frameworks relative to the complexity and value of the commodity chain it covers. The LBMA Responsible Gold Guidance alone cost \$20–40M system-wide. TrueValue's advantage is that it builds on top of existing LBMA infrastructure rather than replacing it — adding the balance-scoring and resource-flow analytical layer that the LBMA framework does not provide.

SECTION 6 — DEPLOYMENT ROADMAP

The Minimum Credible Deployment Path

The roadmap below represents the most efficient realistic path — not the fastest theoretically possible, but the fastest that would produce a Framework with genuine industry validity. Cutting the timeline below these minimums risks producing a scoring system with insufficient data depth to survive regulatory or academic scrutiny.

Month 0

FRAMEWORK DESIGN
BEGINS;
CORE TEAM OF 4–6
HIRED

Month 24

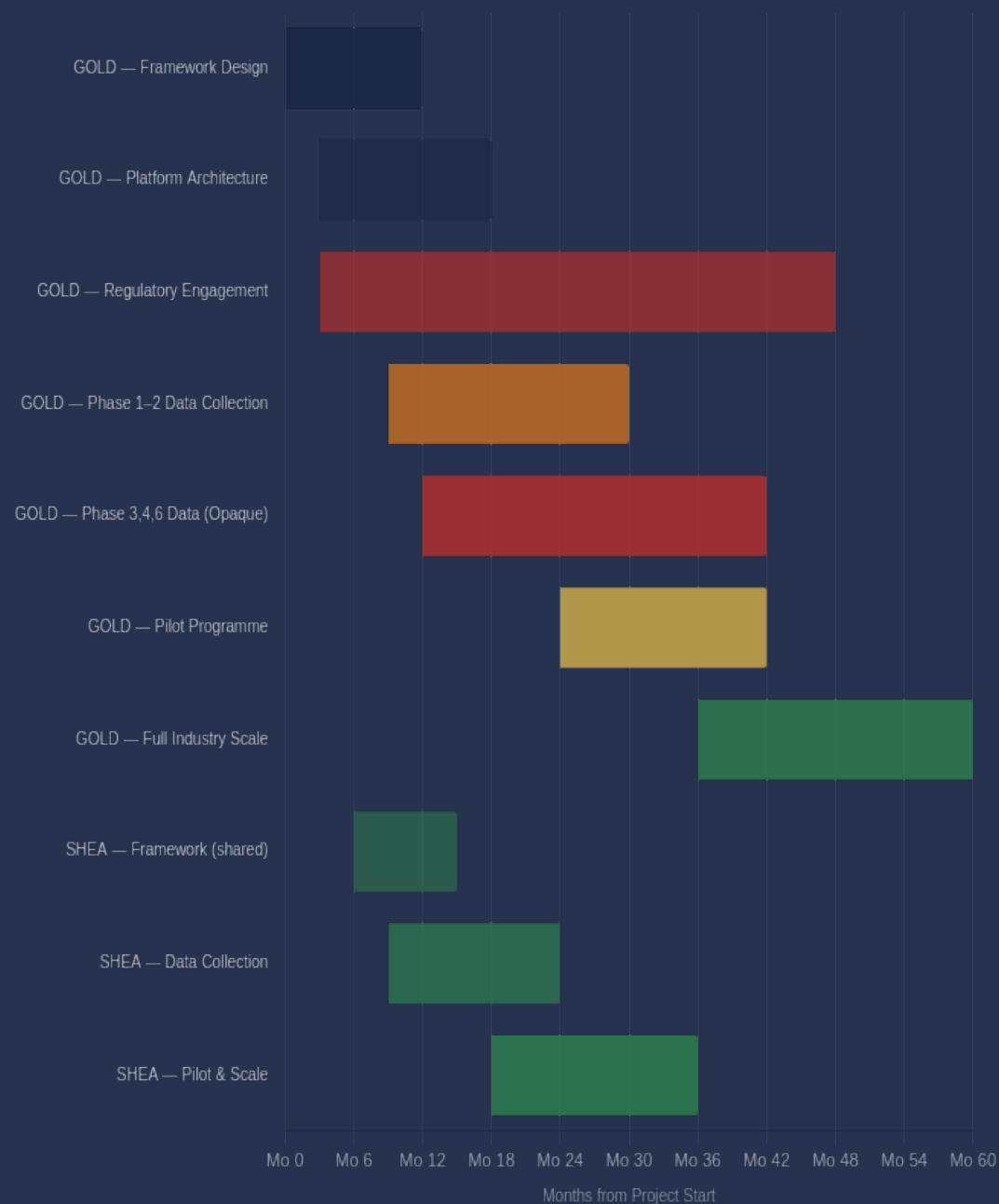
PILOT DATA AVAILABLE;
FIRST BALANCE
SCORES PUBLISHED

Month 48– 60

FULL INDUSTRY
DEPLOYMENT;
STEADY-STATE
OPERATIONS BEGIN

DEPLOYMENT TIMELINE — GOLD & SHEA (PARALLEL)

Horizontal bars show duration of each workstream. Shea runs in parallel from Month 6, sharing technology platform and methodology. Gold's opaque-phase data collection is the critical path item.



THE CRITICAL PATH

The deployment is not constrained by technology or money — it is constrained by data access at three gold supply chain phases: Doré/Intermediate (Phase 3), Refining (Phase 4), and Logistics & Vaulting (Phase 6). These phases are architecturally designed for opacity. Obtaining reliable, auditable data requires either regulatory mandate (fastest path) or extended voluntary negotiation with individual operators (slowest, most expensive path). Every other workstream can run in parallel; these three data negotiations are serial dependencies.

The recommended first action is therefore not a technology build — it is a **regulatory engagement strategy**: approaching LBMA and WGC with a co-development proposal that frames TrueValue as an extension of their existing responsible sourcing infrastructure, not a competitive alternative. This positions the Framework to benefit from mandatory data flows rather than having to negotiate each one individually.

SECTION 7

The Investment Case: Cost vs. Value Unlocked

Implementation costs must be evaluated against the structural value the Framework is designed to unlock. The Investor Report documents a 71% cumulative value advantage for TrueValue-managed chains over a 10-year horizon. Even at conservative estimates of the gold supply chain's annual value (~\$250B), the Framework's optimisation potential — capturing just 1% of currently leaked structural value — generates annual returns of \$2.5B+. The implementation cost of \$13–\$30M is a rounding error relative to that potential.

WHO SHOULD FUND THIS — AND WHY

- **Major mining companies** (co-development): benefit from ESG premium pricing and reduced regulatory risk. Contribution: \$1–3M each for 3–5 anchor participants
- **Institutional investors** (impact/ESG mandates): obtain phase-resolved supply chain risk data not available elsewhere. Contribution: \$2–5M total as consortium
- **LBMA / WGC** (standards body mandate): natural home for ongoing operations; existing regulatory infrastructure reduces deployment friction by 40–60%
- **Development finance (Shea)**: IFC, GIZ, EU grants could cover 30–50% of Shea deployment; Shea becomes proof-of-concept for gold
- **ESG data platforms** (Bloomberg, MSCI, Sustainalytics): TrueValue scores are a premium data product; licensing revenue can self-fund steady-state operations within 3–4 years

WHAT HAPPENS IF IT IS NOT BUILT

- **Continued structural opacity** at Phases 3, 4, and 6 means investors cannot price hidden constraint accumulation — the exact risk profile that generates sudden, correlated supply chain failures
- **ESG capital reallocation pressure** increases without structural data to defend supply chain choices; gold and commodity producers face growing cost-of-capital premium with no measurement tool to respond
- **Regulatory gap**: EU CSRD, SEC climate disclosure, and OECD due diligence requirements are moving toward exactly the kind of phase-level supply chain data TrueValue provides; first movers capture the compliance alignment premium
- **Shea value leakage continues**: \$25–80M/year in unmonetised water and biodiversity assets remains off the balance sheet; women farmers receive below-market compensation with no structural measurement to demonstrate or negotiate against

THE IMPLEMENTATION THESIS IN ONE SENTENCE

At \$13–\$30M over five years for gold, and \$1.8–\$4M over three years for shea, the

TrueValue Framework is not an expensive project — it is a modestly priced structural measurement system for commodity chains that collectively move trillions of dollars of value annually, and whose hidden imbalances represent multiples of the Framework's cost in annually leaking value.

APPENDIX

Methodology & Estimation Assumptions

COST ESTIMATION METHODOLOGY

All cost estimates use a bottom-up labour-hours approach: role types are identified per phase, staffing levels estimated based on analogous project deployments, and blended day rates applied by role (analyst: \$800–\$1,400/day; engineer: \$900–\$1,500/day; field researcher: \$400–\$700/day; legal: \$300–\$600/hr). Ranges reflect the difference between a cooperative industry posture (low) and a resistant one (high).

ANALOGOUS FRAMEWORK SOURCES

- LBMA Responsible Gold Guidance — public annual reports and industry cost surveys
- Fairtrade International — annual report financial statements; academic analysis of standard-setting costs
- CDP — published annual reports and 10-year retrospectives
- GRI — organisational history documentation and budget disclosures
- Acorn Carbon Programme — Serious Shea / Rabobank programme documentation
- Foran, Lenzen & Dey (2005) — TBL methodology precedent; CSIRO operational research

KEY ASSUMPTIONS

- Day rates are in 2026 USD; no forward inflation adjustment applied
- Technology platform assumes 60% SaaS / 40% custom build hybrid
- Contingency of 15% applied to all phases
- Gold pilot covers 2–3 major mine operations, 1 LBMA refinery, and COMEX exchange data
- Shea pilot covers Acorn-enrolled cooperatives in Burkina Faso and Senegal
- Steady-state cost assumes industry cost-sharing via licensing fees from Year 4 onwards

WHAT THIS ESTIMATE DOES NOT COVER

- Policy lobbying costs (if regulatory mandate route pursued)
- Marketing and external communications for Framework adoption
- Third-party academic peer review and journal publication
- Translation and localisation costs for non-English jurisdictions
- Litigation costs above the 15% contingency reserve
- Costs borne by participating entities to integrate with the Framework (estimated separately at \$200k–\$1M per major participant)

TrueValue Analytics — Structural Supply Chain Intelligence. This document presents order-of-magnitude implementation cost estimates based on analogous framework analysis. It does not constitute a project proposal, budget commitment, or financial guarantee. © 2026 TrueValue Analytics. All rights reserved.